

Mid-Term Examination: Coordination Chemistry

Class 12 Board & JEE Level Test (Max Marks: 50, Time: 2 Hours)

Chapter / Focus Area: Inorganic Chemistry - Coordination Compounds

Section A: Conceptual & Objective Questions

- Q1. Write the IUPAC name of $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$ and $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$.
- Q2. What is the magnetic behavior and hybridisation of $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ on the basis of VBT?
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Section B: Analytical & Derivation Questions

- Q3. On the basis of Crystal Field Theory, write the electronic configuration of d^4 in strong field ($\Delta_o > P$) and weak field ($\Delta_o < P$) octahedral complexes.
- Q4. Explain linkage isomerism, ionization isomerism, and coordination isomerism with one example each.
- Q5. Why is $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ colored while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colorless? Explain on the basis of d-d transitions.
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